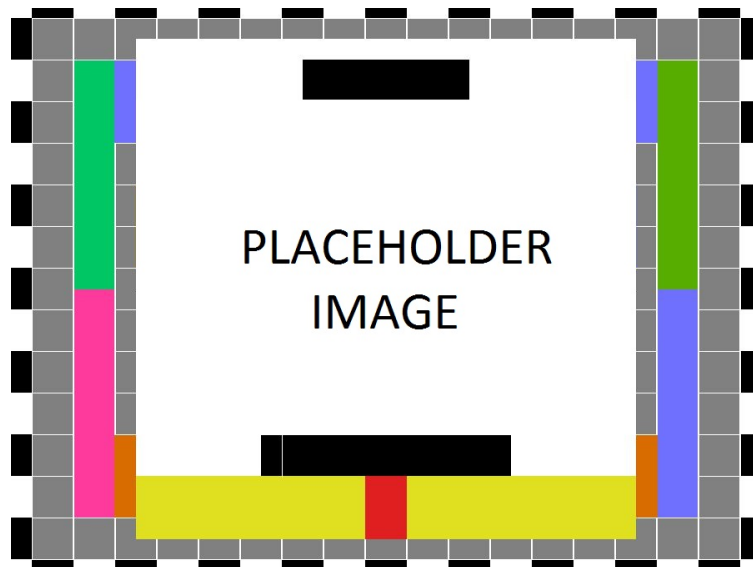


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION  
CSE 4316: SENIOR DESIGN I  
FALL 2026**



**TEAM NIMBUS  
AQUAGUARD SMART WATER-LEAK MONITOR**

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## REVISION HISTORY

| Revision | Date       | Author(s)  | Description       |
|----------|------------|------------|-------------------|
| 0.1      | 10.01.2026 | JVN        | document creation |
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# 1 INTRODUCTION

This SRS specifies AquaGuard Smart Water-Leak Monitor, a BLE leak sensor with a companion mobile app. It follows ISO/IEC/IEEE 29148:2018 [1].

## 1.1 PURPOSE

Agreed requirements baseline for Team Nimbus and its advisor.

## 1.2 SCOPE

The system shall detect moisture/abnormal flow at the device, alert the paired phone locally over BLE, and deliver remote push notifications via a cloud backend when the phone is away; the app shall pair, configure thresholds, and show event history. Whole-home water shut-off actuation is out of scope.

## 1.3 PRODUCT OVERVIEW

### 1.3.1 PRODUCT PERSPECTIVE

Three cooperating elements: the sensor device (BLE peripheral), the mobile app (BLE central), and a lightweight cloud backend (accounts + push).

### 1.3.2 PRODUCT FUNCTIONS

Leak/flow sensing; BLE pairing/config; local + remote alerting; event history; battery/status reporting; firmware update (OTA, stretch).

### 1.3.3 USER CHARACTERISTICS

A general homeowner using a personal smartphone; no technical expertise assumed.

### 1.3.4 ASSUMPTIONS & DEPENDENCIES

Recent iOS/Android phones; BLE range  $\leq 5$  m; free-tier push service; controlled water test rig.

## 1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

| Term               | Definition                                                   |
|--------------------|--------------------------------------------------------------|
| BLE                | Bluetooth Low Energy                                         |
| Peripheral/Central | BLE device roles (device advertises; phone connects)         |
| GATT               | Generic Attribute Profile (BLE service/characteristic model) |
| Leak event         | A moisture/flow reading crossing the configured threshold    |

## 2 REQUIREMENTS ENGINEERING APPROACH

Identifiers [TYPE] - [NN] (EI, FR, US, PE, SA, SE, MS, QA) with the 29148 attribute set. Requirements are singular and verifiable.

## **3 EXTERNAL INTERFACE REQUIREMENTS**

### **3.1 EI-01: BLE GATT PROFILE**

#### **DESCRIPTION**

The device shall expose a BLE GATT service with characteristics for live reading, leak-event notification, threshold configuration, and battery level.

#### **RATIONALE**

The app and device must share a well-defined wireless contract.

#### **SOURCE**

Team.

#### **PRIORITY**

Critical

#### **VERIFICATION METHOD**

Test — read/write/subscribe each characteristic from the app.

#### **CONSTRAINTS**

Bluetooth SIG GATT constraints; MTU limits.

#### **APPLICABLE STANDARDS**

Bluetooth Core Specification (BLE).

### **3.2 EI-02: PUSH NOTIFICATION SERVICE**

#### **DESCRIPTION**

The backend shall deliver leak alerts to the app via the platform push services (APNs/FCM).

#### **RATIONALE**

Alerts must reach the user when the app is backgrounded or the phone is away.

#### **SOURCE**

Team.

#### **PRIORITY**

High

#### **VERIFICATION METHOD**

Test — trigger a leak; receive a push with app backgrounded.

#### **CONSTRAINTS**

Depends on APNs/FCM availability.

#### **APPLICABLE STANDARDS**

APNs/FCM specifications.

## **4 FUNCTIONAL REQUIREMENTS**

### **4.1 FR-01: LEAK DETECTION**

#### **DESCRIPTION**

The device shall raise a leak event within 5 s of moisture or abnormal flow crossing the configured threshold.

#### **RATIONALE**

Early detection is the product's core purpose.

#### **SOURCE**

Advisor; team.

#### **PRIORITY**

Critical

#### **VERIFICATION METHOD**

Test — apply water on the rig; measure detection time.

#### **CONSTRAINTS**

Threshold configurable via EI-01.

#### **APPLICABLE STANDARDS**

None.

### **4.2 FR-02: BLE PAIRING & CONFIGURATION**

#### **DESCRIPTION**

A user shall pair the device with the app and set the leak threshold through the app UI.

#### **RATIONALE**

The device must be set up without special tools.

#### **SOURCE**

Advisor.

#### **PRIORITY**

Critical

#### **VERIFICATION METHOD**

Demonstration — pair and set a threshold.

#### **CONSTRAINTS**

One device per app instance in the prototype.

#### **APPLICABLE STANDARDS**

None.

### **4.3 FR-03: LOCAL ALERT**

#### **DESCRIPTION**

On a leak event while connected, the app shall alert the user within 3 s with a distinct visual and audible/haptic notification.

#### **RATIONALE**

Immediate on-phone warning when the user is nearby.

#### **SOURCE**

Advisor.

#### **PRIORITY**

Critical

#### **VERIFICATION METHOD**

Test — trigger event; measure alert latency.

#### **CONSTRAINTS**

Requires an active BLE connection.

#### **APPLICABLE STANDARDS**

None.

### **4.4 FR-04: REMOTE ALERT**

#### **DESCRIPTION**

When the phone is not connected to the device, a leak event shall be delivered as a remote push notification within 30 s of reconnection/relay.

#### **RATIONALE**

Users are often away from the sensor; alerts must still reach them.

#### **SOURCE**

Advisor.

#### **PRIORITY**

High

#### **VERIFICATION METHOD**

Test — background the app; trigger event; receive push.

#### **CONSTRAINTS**

Depends on connectivity and EI-02.

#### **APPLICABLE STANDARDS**

None.

### **4.5 FR-05: EVENT HISTORY**

#### **DESCRIPTION**

The app shall display a timestamped history of leak events and battery status for the paired device.

#### **RATIONALE**

Users need to review and trust past behavior.

#### **SOURCE**

Advisor.

#### **PRIORITY**

Moderate

#### **VERIFICATION METHOD**

Demonstration — view history after several events.

#### **CONSTRAINTS**

History retained locally and in the backend.

#### **APPLICABLE STANDARDS**

None.

## **5 USABILITY REQUIREMENTS**

### **5.1 US-01: GUIDED SETUP**

#### **DESCRIPTION**

A first-time user shall complete pairing and threshold setup in under 5 minutes following the in-app guide in  $\geq 8$  of 10 trials.

#### **RATIONALE**

Consumer devices must be self-installable.

#### **SOURCE**

Advisor.

#### **PRIORITY**

High

#### **VERIFICATION METHOD**

Test — moderated setup trials.

#### **CONSTRAINTS**

On both iOS and Android.

#### **APPLICABLE STANDARDS**

WCAG 2.2 AA (mobile).

## **6 PERFORMANCE REQUIREMENTS**

### **6.1 PE-01: BATTERY LIFE**

#### **DESCRIPTION**

The device shall operate at least 6 months on one battery set under nominal duty (advertising + occasional connection).

#### **RATIONALE**

Frequent battery changes would make the product impractical.

#### **SOURCE**

Advisor.

#### **PRIORITY**

High

#### **VERIFICATION METHOD**

Analysis — power budget + accelerated measurement.

#### **CONSTRAINTS**

Coin/AA battery capacity.

#### **APPLICABLE STANDARDS**

None.

### **6.2 PE-02: DETECTION RELIABILITY**

#### **DESCRIPTION**

On the controlled test rig, leak detection shall achieve  $\geq 95\%$  true-positive rate with  $\leq 1$  false alarm per 24 h idle.

#### **RATIONALE**

Missed leaks defeat the purpose; nuisance alarms erode trust.

#### **SOURCE**

Advisor; team.

#### **PRIORITY**

High

#### **VERIFICATION METHOD**

Test — repeated leak/idle trials on the rig.

#### **CONSTRAINTS**

Limited to tested leak scenarios.

#### **APPLICABLE STANDARDS**

None.

## **7 SAFETY REQUIREMENTS**

### **7.1 SA-01: ELECTRICAL & WATER SAFETY**

#### **DESCRIPTION**

The device shall isolate its electronics from the moisture pad and operate at safe low voltage (battery); no mains connection.

#### **RATIONALE**

Placing electronics near water demands isolation to avoid hazard.

**SOURCE**

Team; safety review.

**PRIORITY**

Critical

**VERIFICATION METHOD**

Inspection — enclosure/isolation review.

**CONSTRAINTS**

Battery-only design (charter constraint).

**APPLICABLE STANDARDS**

IEC 60950/62368 (guidance).

## **8 SECURITY & PRIVACY REQUIREMENTS**

### **8.1 SE-01: SECURE PAIRING & TRANSPORT**

**DESCRIPTION**

BLE pairing shall use a bonded, encrypted link, and app↔backend traffic shall use TLS 1.3.

**RATIONALE**

Prevents spoofed devices and interception of alerts/account data.

**SOURCE**

Team; security review.

**PRIORITY**

Critical

**VERIFICATION METHOD**

Test — verify BLE bonding; TLS scan.

**CONSTRAINTS**

BLE security modes supported by the module.

**APPLICABLE STANDARDS**

Bluetooth LE Security; RFC 8446.

## **9 MAINTENANCE & SUPPORT REQUIREMENTS**

### **9.1 MS-01: DIAGNOSTICS**

**DESCRIPTION**

The app shall surface device firmware version, battery level, and last-seen time to aid support.

**RATIONALE**

Enables remote troubleshooting.

**SOURCE**

Team.

## PRIORITY

Moderate

## VERIFICATION METHOD

Demonstration.

## CONSTRAINTS

Fields exposed via EI-01.

## APPLICABLE STANDARDS

None.

# 10 OTHER QUALITY ATTRIBUTES

## 10.1 QA-01: CROSS-PLATFORM PARITY

### DESCRIPTION

Core functions (pair, configure, local alert, history) shall behave equivalently on iOS and Android.

### RATIONALE

Users on either platform must get the same protection.

### SOURCE

Advisor.

## PRIORITY

High

## VERIFICATION METHOD

Test — run the core test suite on both platforms.

## CONSTRAINTS

Platform BLE-background differences documented.

## APPLICABLE STANDARDS

ISO/IEC 25010:2023 (portability) [2].

# 11 VERIFICATION & TRACEABILITY

| Req ID | Requirement (abbrev.)     | Method | Acceptance                    |
|--------|---------------------------|--------|-------------------------------|
| FR-01  | Leak detection $\leq 5$ s | T      | measured $\leq 5$ s           |
| FR-03  | Local alert $\leq 3$ s    | T      | measured $\leq 3$ s           |
| FR-04  | Remote push               | T      | push received<br>backgrounded |
| PE-01  | 6-month battery           | A      | power budget<br>meets target  |
| SE-01  | Secure pairing/TLS        | T      | bonded BLE;<br>TLS1.3         |

## **12 FUTURE ITEMS**

### **12.1 FI-01: AUTOMATIC WATER SHUT-OFF**

#### **DESCRIPTION**

An add-on valve actuator that shuts off water on a confirmed leak.

#### **RATIONALE**

High value but exceeds prototype scope and budget.

#### **SOURCE**

Advisor.

#### **PRIORITY**

Future

#### **VERIFICATION METHOD**

Test (if implemented).

#### **CONSTRAINTS**

Requires plumbing actuator + safety work.

#### **APPLICABLE STANDARDS**

None.

#### **REFERENCES**

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2018.
- [2] ISO/IEC, “ISO/IEC 25010:2023 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.